Co-Existence Analysis of Terrestrial and Non-Terrestrial Networks in S-Band Using Stochastic Geometry

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***Abstract*—Co-existence of terrestrial and non-terrestrial networks (NTN) is foreseen as an important component to fulfill the global coverage promised for sixth-generation (6G) of cellular networks. Due to ever-rising spectrum demand, using dedicated frequency bands for terrestrial network (TN) and NTN may not be feasible. As a result, certain S-band frequency bands allocated by radio regulations to NTN networks overlap with those already utilized by cellular TN, leading to significant performance degradation due to the potential co-channel interference. Early simulation-based studies on different co-existence scenarios failed to offer a comprehensive and insightful understanding of the overall performance of these networks. Besides, the complexity of a brute force performance evaluation increases exponentially with the number of nodes and their possible combinations in the network. In this paper, we utilize stochastic geometry to analytically derive the performance of TN-NTN integrated networks in terms of the probability of coverage and average achievable data rate for two co-existence scenarios. From the numerical results, it can be observed that, depending on the network parameters, TN and NTN users' distributions, and traffic load, one co-existence case may outperform the other, resulting in the optimal performance of the integrated network. The analytical results presented herein pave the way for designing state-of-the-art methods for spectrum sharing between TN and NTN and optimizing the integrated network performance.**

***Index Terms*—Non-terrestrial networks (NTN), low earth orbit (LEO) internet constellations, interference, co-existence, stochastic geometry, coverage probability, data rate, spectrum sharing.**

I. Introduction

NON-TERRESTRIAL networks (NTN) are foreseen as an integral part of sixth-generation (6G) cellular networks to fulfill the requirements on expanding coverage and connectivity in remote and underserved areas [1]. NTN encompass different types of networks communicating through the sky (air or space), including satellite-based networks, high-altitude platform systems (HAPSs), and unmanned aerial vehicles (UAVs). During recent years, due to the significant reduction in the launch costs of satellites and increased demand for global broadband connectivity, low Earth orbit (LEO) satellites outpaced other types of NTN in commercialization, leading to several initiatives such as Starlink, OneWeb, AST SpaceMobile and Project Kuiper [2].

The integration of NTN technology into terrestrial networks (TN) for the next-generation communication systems, such as 6G, has great potential for revolutionizing global connectivity [3], by complementing terrestrial infrastructure and extending connectivity to remote areas, such as rural regions, maritime environments, and disaster-stricken locations. However, the successful deployment and operation of NTN systems face several challenges, one of which is the scarcity of spectrum resources. Due to limited availability of frequency bands, especially in lower frequency bands like S-band, the International Telecommunication Union (ITU) Radio Regulations (RR) allocated radio frequency ranges to NTN, n255 and n256, which is already occupied, at least partly, by fifth-generation New Radio (5G NR) network [4]. As a result, co-channel interference is inevitable between both networks. To address this limitation, it becomes imperative to explore mechanisms for sharing and co-existence between NTN and TN, while ensuring efficient spectrum utilization and minimizing interference.

Due to the massive number of nodes and the large number of possible combinations, Monte Carlo simulation-based performance evaluation of the TN-NTN integrated network is computationally demanding. In this paper, to evaluate the interference effect of an NTN network on the performance of a TN network, we utilized stochastic geometry as a powerful mathematical framework which has a rich history in modeling and analyzing of wireless networks [5]. Stochastic geometry enables obtaining analytical tractable derivations on the performance of the integrated network using the spatial distribution of network elements, e.g., base stations (BSs) and user equipments (UEs). The analysis provides several insights into the impact of different network parameters, such as the density of TN BSs, NTN BSs' altitudes, inter-site distance (ISD), the number of UEs, and the minimum separation distance between users of each network, on the network's performance. Such results will pave the way for the design of different co-existence mechanisms and spectrum sharing policies required to enable seamless operation of NTN and TN systems.

A. Related Works

The literature on TN and NTN co-existence is gaining momentum due to the emergence of LEO constellations. In

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[6], the performance of 5G TN is compared with that of a LEO satellite network using an experimental setup in a suburban environment, in terms of latency and throughput. It was demonstrated that multi-connectivity between both networks can provide coverage seamlessly for low-latency and high-reliability requiring services. Co-existence and spectrum sharing studies have received significant attention in standardization activities. For instance, a simulation tool that analyzes aggregate interference between 5G TN and other networks following ITU recommendations was presented in [7]. It was used, for instance, in [8] to investigate co-existence of 5G and HAPS.

Several methods for spectrum sharing between TN and LEO satellites have also been suggested recently. In [9], a methodology using open radio-access network (ORAN) was proposed, in which interference is reported by the ORAN terrestrial networks to the satellite service provider. 3rd Generation Partnership Project (3GPP) launched several standardization activities through release 15 to 18 to support integration of satellite networks and 5G NR. The flexible spectrum sharing is among the topics identified for immediate and longer-term commercial needs [10], [11]. Several co-existence scenarios have been identified in [4]. Despite being highly significant in standardization activities, the scientific literature failed to provide a unified framework to analyze the performance of these co-existence scenarios [12], [13] at the system-level. This is because of over simplifying the system models or considering only very specific cases. For instance, the results of the co-existence study in [13] are limited to considering only one interfering node.

The simulation-based study on spectrum sharing between TN and NTN was conducted in [14] and [15] under different TN-NTN co-existence scenarios, including direct pairing and reverse pairing, where TN and NTN operate over the same and opposite directions, respectively. The studies aim to enhance NTN uplink operations by mitigating co-channel interference. An experimental-based study on spectrum sharing between TN and a LEO satellite network was presented in [16], with the objective of minimizing the interference from satellites at the TN users. Cognitive radio approaches for co-existence of satellites and TN cellular networks over the same frequency bands have been studied in [17] and [18], targeting better exploitation of frequencies that are not used by the terrestrial network. An overview of the integration of both geostationary earth orbit (GEO) and LEO satellites with 5G systems, focusing on enhanced mobile broadband (eMBB) and narrowband-IoT (NB-IoT) was presented in [19]. The above literature on spectrum sharing between TN and NTN indicates that the results are based on brute force simulations, which fail to conclude the generic performance of the integrated network in terms of both networks' parameters.

Stochastic geometry is a mathematical tool that finds extensive application across various scientific disciplines including wireless communication. Its utilization in the analysis of wireless networks dates back to 1961 to investigate the connectivity of large wireless networks with huge number of nodes [20]. Stochastic geometry is utilized for system-level performance analysis of large wireless networks to avoid massive time consuming Monte Carlo simulations. Using stochastic geometry, the performance of the network is obtained by taking the averages over different realizations of nodes in a network [5]. Averages can be computed over a large number of nodes' locations or several network realizations, for different metrics of interest, e.g., the probability of coverage and the average achievable rate. More particularly, the nodes' locations are modeled as a point process which captures the characteristics of their distribution, e.g., their density and the relative distances between them.

Stochastic geometry has found an extensive application in the analysis of planar two-dimensional networks including heterogeneous [21], [22], [23], [24], [25], cognitive [26], [27], ad hoc [28], [29], [30], [31], and vehicular [32], [33] networks. A comprehensive literature survey on the utilization of stochastic geometry for modeling multi-tier and cognitive cellular networks has been provided in [24]. A taxonomy is founded on three key criteria: the type of network under consideration, the point process employed for modeling, and the network performance characterization techniques. The coverage and rate are tractably derived through modeling the BSs as a homogeneous Poisson point process (PPP) in [34] which is known as one of the seminal works on the utilization of stochastic geometry for performance analysis of a terrestrial cellular. The model shows the optimistic results of conventional grid models while the stochastic model provides a lower band on the network performance.

The utilization of stochastic geometry to analyze three-dimensional networks has been also compelling [35], [36], [37]. The tool was used to study different scenarios such as BSs being located on both ground and rooftops in highly crowded urban areas, UAV swarming, and massive satellite networks. Two spectrum sharing techniques, i.e., underlay and overlay, to share the spectrum between UAV-to-UAV and BS-to-UAV transceivers are analyzed in [38] by modeling both UAVs and BSs as homogeneous PPPs. The impact of interference from TN BSs on the primary users of a multi-beam geostationary satellite when both networks share the same spectrum resources is investigated in [39]. Using tools from stochastic geometry, authors analyzed two performance metrics of outage probability and area spectral efficiency under three secondary transmission schemes by modeling the satellite users and BSs as two independent point processes.

The application of stochastic geometry has been also extended for performance analysis of massive low Earth orbit constellations, by modeling the satellites as a binomial point process (BPP) [40], [41], [42], [43] or a nonhomogeneous PPP [44], [45] on a spherical shell which facilitates the utilization of the tools from stochastic geometry. In fact, a point process represents different realizations of satellites over the time caused by the motion of satellites on the orbits. Verification of the exact analytical derivations from stochastic modeling with the actual simulated constellations represents a fair accuracy of such modeling. In [46], the coverage and rate of a noise-limited scenario is studied using stochastic geometry when the user is associated with a satellite that provides the highest signal-to-noise ratio (SNR). Such association, compared to the simplistic case of associating the user with the nearest satellites, includes the effect of shadowing caused by the user's surrounding objects. The performance of multi-altitude LEO constellations when satellites are distributed on several orbital shells is studied in [42] and [47].

In this paper, we study the coverage probability and average data rate of a primary TN network that shares the same spectrum resources with a secondary LEO network in the S-band. We utilize tools from stochastic geometry to investigate two scenarios in which the TN downlink (DL) transmission is subject to interference from either satellites or NTN UEs transmitting in uplink (UL). Using the tools from stochastic geometry facilitates the derivation of analytical exact expressions for the performance of the integrated TN-NTN network. In contrast to [48], which mainly quantifies interference from the terrestrial network toward satellite receivers, our focus is on the performance of terrestrial networks under NTN interference and on identifying spectrum arrangements that enable efficient coexistence; in this sense, [48] is complementary to our work. The results herein provide direct insights into several parameters associated with the integrated network, such as the satellite altitude, the TN user location, transmission power of BSs and satellites, traffic load, and d_{ISD}.

B. Contributions and Paper Organization

In this paper, we consider a group of TN BSs distributed over a finite region that is also under the coverage of a beam from a LEO satellite. The TN and LEO networks share the same frequency channels in the S-band. The UEs, located further away from the TN network's coverage area, lack access to the service provided by the TN network and, thus, are served by the LEO network. In order to analyze the performance of the integrated network and utilize the tools from stochastic geometry, we model the TN BSs and the NTN UEs as a BPP, which is a popular model analogous to PPP. Using BPP is the most suitable choice to study co-existence cases in this paper as a given number of nodes is deployed over a given finite region [49], bounded by the satellite's beam coverage. Another reason that justifies the application of BPP is that the user experience varies with location [50], whereas for PPP, the performance is independent of the actual locations of the nodes. For example, as we will show in the numerical results provided in Section IV, the aggregate interference is different for a user in the center of the cell than that of network edge. It is worth noting that a Matérn or Ginibre process will give similar trends at the cost of more algebraic complexity.

The main scientific contributions of this paper are as follows:

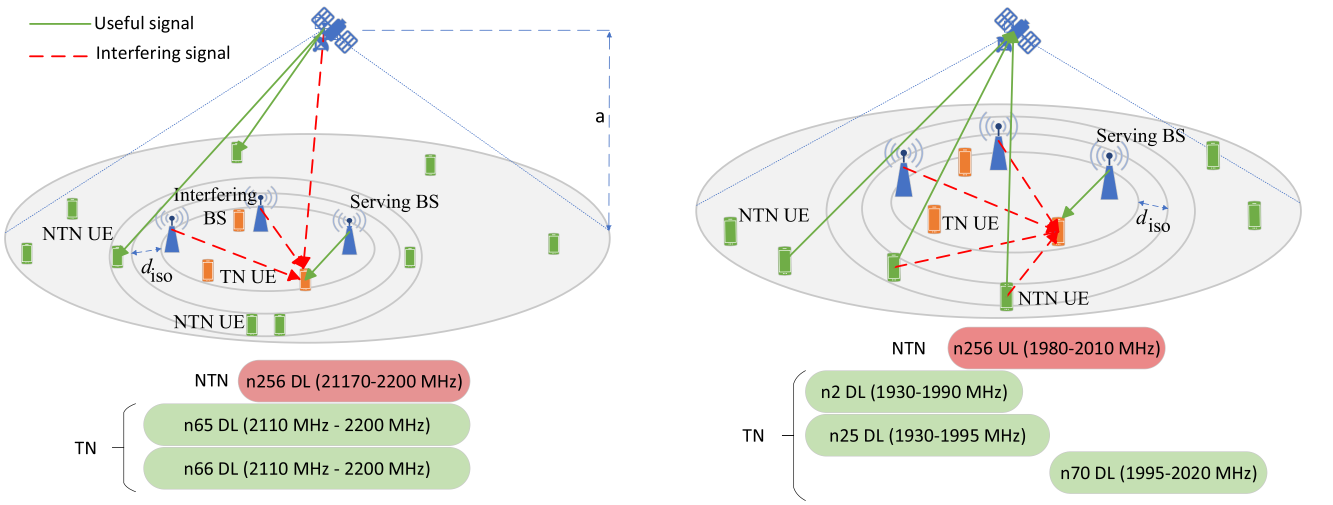
- Due to overlap between the frequency range allocated to NTN and TN, co-existence of the two networks may result in co-channel interference. In this paper, performance analysis is provided for two co-existence cases from [4], which will be referred to as Case I and Case II in what follows. Case I corresponds to the case when TN and NTN networks communicate simultaneously with their corresponding users over the same frequency division duplex (FDD) DL bands, i.e., the source of interference is from the satellite that transmits the signal to the area where the TN network is located. In Case II, the TN network's FDD DL bands are shared with FDD UL frequency band of NTN, i.e., the source of interference is from the NTN UEs transmitting to their serving satellite in close proximity of the TN network. Note that according to [12], the performance degradation in case of co-existence of TN UL with NTN network is insignificant.
- Using the tools from stochastic geometry, we derive exact analytical expressions for the probability of coverage and average achievable data rate of the above-mentioned co-existence scenarios.
- We validate all our theoretical derivations with Monte Carlo simulations. Throughout the numerical results, we show how each of the co-existence cases may affect the performance of TN network. The performance is illustrated in terms of several parameters, e.g., TN user's location in the cell, type of geographical environments, satellite's altitude, TN BS's transmit power, the number of NTN users, TN traffic load, BSs' density, and the isolation distance between TN and NTN UEs.
- The numerical results suggest several insightful guidelines on spectrum sharing between TN and NTN networks. In other words, the results provide the answer to the following question: *Given the network deployment parameters, user's distributions, and traffic load, which case of co-existence leads to a superior performance for the TN network?*

For the propagation model, we consider Nakagami-m fading with integer m that not only preserves analytical tractability but also covers a wide range of fading cases by varying the parameter m. As increasing m corresponds to higher probability of line-of-sight, we opt for a higher m for satellite to ground channels, while a smaller value is used for terrestrial channels. From the numerical results, we will see that there is less performance degradation in urban regions when TN DL shares the same band with NTN DL. In the other scenario, when NTN UL uses the same band as TN DL, the performance varies significantly with the TN user's location w.r.t. the cell center, i.e., more degradation occurs for edge users.

The remainder of this paper is organized as follows. Section II describes the system model as well as the two TN-NTN co-existence scenarios studied in this paper. Then in Section III, we derive the exact analytical expressions for the coverage performance and the average data rate for each co-existence case. Verification of the analytical results as well as studying the effect of several network parameters on the performance of the integrated network, is provided in Section IV. Finally, the paper is concluded in Section V.

II. SYSTEM MODEL

The TN-NTN co-existence cases studied in this paper are shown in Fig. 1. In these cases, a LEO satellite orbiting at altitude a serves N_{u} NTN users in DL (c.f. Fig. 1(a)) and UL (c.f. Fig. 1(b)). For both cases, a TN network communicates with its UEs in the DL direction over the same FDD bands allocated to the NTN network. The frequency bands allocated



(a) NTN DL interferes with TN DL.

(b) NTN UL interferes with TN DL.

Fig. 1. Co-existence scenarios from [4] where NTN DL (a) or NTN UL (b) interferes with TN DL.

to NTN by ITU-R with the overlapping 5G NR bands are depicted in Fig. 1 as well. It is worth noting that the overlap ratio of the overall bands does not affect the per unit frequency interference analysis considered in this paper. Assuming that the TN network is always preferable over NTN due to its superior performance and reliability, NTN UEs are assumed to be distant from the TN network's edge at least by a distance denoted by $r_{\rm iso}$. This isolation distance reflects NTN deployment in areas sparsely covered by TN, as described in [4] and ITU guidelines. The TN BSs are distributed as clusters with radius $r_{\rm TN}$, within the beam coverage area of the satellite, with each cluster having $N_{\rm c}$ BSs and the minimum distance between any two BSs, known as inter-site distance, is denoted by $d_{\rm ISD}$, which is a function of the physical environment, e.g., rural and urban. To softly take into account the repulsion between the TN BSs without losing tractability, $r_{\rm TN}$ is chosen to be linearly proportional to $d_{\rm ISD}$. Terrestrial UEs are uniformly distributed within each TN cluster.

In this paper, the serving TN BS is assumed to be the nearest BS to the TN user and serves the TN UE in the DL direction. We assume that all TN BSs on the same cluster and the NTN share the same frequency band. This causes co-channel interference at the TN UE's reception. In Fig. 1, solid lines represent useful signals, while dashed lines represent the interfering signals.

The channel model for serving link follows a Nakagami-m fading model which allows to account for a wide range of multi-path fading conditions by varying the parameter m, while preserving the analytical tractability. Following the same approach as in [40] and [44], we do not need to limit the fading distribution of the interfering channels to any specific one for analytical derivations, since it has no effect on the tractability of our analysis. Obviously, resorting to some specific channel models is required to generate the numerical results. Since the m parameter is an indicator of the fading severity, a larger value should be opted for satellite to ground channel compared to terrestrial links with less line-of-sight probability. The channel gains for TN and NTN links are denoted by $H_{\rm TN}$ and $H_{\rm NTN}$, respectively.

The antennas on the satellites and TN BSs are directional and the beamforming patterns are chosen according to the specifications given in [4]. The antennas can radiate multiple beams towards the ground. The TN and NTN users are equipped with omni-directional antenna elements [4], [51]. Assuming perfect beamforming, the overall maximum antenna gain will be expressed as $G_{\rm NTN} = G_{\rm sat}G_{\rm u}$ and $G_{\rm TN} = G_{\rm BS}G_{\rm u}$, where $G_{\rm sat}$, $G_{\rm BS}$, and $G_{\rm u}$ are the maximum antenna gain of satellites, the TN BSs, and the user, respectively.

We study the TN-NTN co-existence under two different scenarios as described in the following subsections. In these scenarios, we study the performance degradation caused by spectrum sharing between TN and NTN networks at the TN UE during NTN network transmission in DL and UL. The analysis is performed per Hz of bandwidth, meaning that SINR is evaluated per unit frequency. Thus, the results are independent of any specific subcarrier spacing. Based on the described scenarios, the signal-to-interference-plus-noise ratio (SINR) at the TN UE can be expressed as

$$\mathrm{SINR}_{\rm TN} = \frac{p_{\rm TN}G_{\rm TN}H_{\rm TN}R_0^{-\alpha_{\rm TN}}}{I_{\rm TN} + I_{\rm NTN} + \sigma^2}, \tag{1}$$

where constant σ^2 is the power of additive thermal noise, the parameter $\alpha_{\rm TN}$ is a path loss exponent for TN link, $p_{\rm TN}$ is the transmit power of TN BS, $I_{\rm TN} = \sum_{n=1}^{N_{\rm c}-1} p_{\rm TN}G_{\rm TN}H_{\rm TN}R_n^{-\alpha_{\rm TN}}$ is the cumulative interference power from all other BSs in the same cluster as the TN user, and $I_{\rm NTN} \in \{I_{\rm DL}, I_{\rm UL}\}$ is the interference power received from the NTN network in DL and UL directions denoted by $I_{\rm DL}$ and $I_{\rm UL}$, respectively. Since the terrestrial cluster radius is much smaller than the satellite footprint, interference from neighboring TN clusters is negligible compared to intra-cluster interference. The distances from the TN user to the serving TN BS and other interfering BSs are denoted by random variables R_0 and R_n, $n = 1, 2, \ldots, N_{\rm c}-1$, respectively. In the following subsections, we elaborate on two TN-NTN co-existence cases studied in this paper.

A. Co-Existence Case I: NTN DL as the Aggressor and TN DL as the Victim

Co-existence Case I corresponds to Scenario 3 given in [4], in which both TN and NTN operate in DL over the same frequency band (c.f. Fig. 1(a)). For instance, NTN satellites might operate over FDD DL n265 band, which could cause interference with the FDD n65 DL and FDD n66 DL bands used by the terrestrial network. The aggressor in this case is a LEO satellite at altitude a and the victim is a TN user, which receives the useful DL signal from a BS. The source of interference at the TN UE will be from other BSs located in the cluster of the serving TN BS, as well as the LEO satellite that transmits towards the TN cluster.

We assume that BSs are distributed according to a BPP on a circle with radius $r_{\rm TN}$, and that the victim TN UE, without loss of generality, is located on point $(x_0, 0, 0)$. Now let us define a circle centered at UE's location with radius equal to the distance between the UE and any BS in the TN cluster, denoted by R_n. The area of this circle is given by

$$\mathcal{A}(r_n) = \begin{cases} \pi r_n^2, & 0 \le r_n \le r_{\rm TN} - x_0 \\ r_n^2\left(\theta - \frac{1}{2}\sin(2\theta)\right) + r_{\rm TN}^2\left(\phi - \frac{1}{2}\sin(2\phi)\right), & r_{\rm TN} - x_0 < r_n < r_{\rm TN} + x_0, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

where $\theta = \arccos\left(\frac{r_n^2 + x_0^2 - r_{\rm TN}^2}{2x_0 r_n}\right)$ and $\phi = \arccos\left(\frac{-r_n^2 + x_0^2 + r_{\rm TN}^2}{2x_0 r_{\rm TN}}\right)$. Thus, for BPP-distributed BSs, the cumulative distribution function (CDF) of the distance between the TN UE and any arbitrary BS is $F_{R_n}(r_n) = \frac{\mathcal{A}(r_n)}{\pi r_{\rm TN}^2}$. Due to the channel assignment by which the serving BS is the nearest one among all the $N_{\rm c}$ i.i.d. BSs, the CDF of R_0 for a cluster of BPP-distributed gNBs can be written as (3), as shown at the bottom of the page.

When conditioned on the serving distance such that $R_0 = r_0$, the CDF of $R_n|_{R_0 = r_0}$, i.e., the distance between the UE and interfering TN BSs, is obtained by conditioning R_n on R_0 as follows:

$$F_{R_n|R_0}(r_n|r_0) \triangleq \mathbb{P}\left(R_n < r_n | R_0 = r_0\right) = \frac{\mathbb{P}\left(r_0 \le R_n \le r_n\right)}{\mathbb{P}(R_n > r_0)} = \frac{F_{R_n}(r_n) - F_{R_n}(r_0)}{1 - F_{R_n}(r_0)}. \tag{4}$$

Taking the derivative w.r.t. r_n gives the probability density function (PDF) of R_n as

$$f_{R_n|R_0}(r_n|r_0) = \frac{f_{R_n}(r_n)}{1 - F_R(r_0)}. \tag{5}$$

The interference power received from a satellite that transmits towards ground users, i.e., $I_{\rm NTN}$ in (1), can be obtained as $I_{\rm DL} = p_{\rm NTN} G_{\rm NTN} H_{\rm NTN} \mathscr{R}_n^{-\alpha_{\rm NTN}}$, where $\mathscr{R}_n$ is the distance between the LEO satellite and the ground users, and $\alpha_{\rm NTN}$ is a path loss exponent for NTN link.

B. Co-Existence Case II: NTN UL as the Aggressor and TN DL as the Victim

Co-existence Case II corresponds to Scenario 5 given in [4] in which TN DL and NTN UL send the data over the same frequency band (c.f. Fig. 1(b)). For instance, NTN UEs might send data over the FDD UL n256 band, which could cause co-channel interference to TN UEs receiving data over FDD DL bands such as n2, n24, and n70. The aggressors in this case are the NTN UEs and the victim, similarly to Case I, is the TN UE, which receives the useful signal from its nearest TN BS. Thus, the source of interference at the TN UE will be from other NTN UEs, surrounding the TN cluster, as well as other BSs sharing the same cluster with the serving BS. Therefore, the distribution of serving distance and TN interfering distances will be similar to Case I as given in (3) and (4), respectively. The NTN interference power can be written as $I_{\rm UL} = \sum_{n=1}^{N_{\rm u}} p_{\rm NTN} G_{\rm NTN} H_{\rm NTN} \mathcal{R}_n^{-\alpha_{\rm TN}}$. To obtain the distribution of the distances between the TN UE and NTN UEs, we assume that NTN UEs are distributed according to a BPP on the annulus between two cocentric circles with radii $r_{\rm TN} + r_{\rm iso}$ and $r_{\rm NTN}$ where $r_{\rm TN} + r_{\rm iso} < r_{\rm NTN}$ and $r_{\rm iso} \ge 0$, as depicted in Fig. 2. Obviously, in this case, the interfering distances are independent of the serving distance. Thus, the distribution is not conditional on R_0. The distances from the TN UE to any NTN UE is denoted by $\mathcal{R}_n$ to distinguish it from R_n, the distances between TN BSs and TN UE. The CDF of $\mathcal{R}_n(r_n)$ is obtained as the ratio of the shaded region area shown in Fig 2 to the total surface area of the annulus where the NTN UEs are distributed as a BPP. The shaded area is the area obtained from the intersection of the disc centered at TN UE with radius $\mathcal{R}_n$ and the outer annulus.

The CDF $F_{\mathcal{R}_n}(r_n) = \frac{\mathcal{A}(r_n)}{\pi(r_{\rm NTN}^2 - (r_{\rm TN} + r_{\rm iso})^2)}$ is a piece-wise function given in (6), where $\mathcal{A}(r_n)$ is the area of shaded region, $w = r_{\rm TN} + r_{\rm iso}$, $F_{a,b}(y) = (y-b)\sqrt{a^2 - (b-y)^2} +$

$$F_{R_0}(r_0) \triangleq \mathbb{P}\left(R_0 \le r_0\right) = 1 - \left(1 - F_R(r_0)\right)^{N_{\rm c}}$$
$$= \begin{cases} 1 - \left(1 - \left(\frac{r_0}{r_{\rm TN}}\right)^2\right)^{N_{\rm c}}, & 0 \le r_0 \le r_{\rm TN} - x_0 \\ 1 - \left(1 - \frac{1}{\pi}\left(\left(\frac{r_0}{r_{\rm TN}}\right)^2\left(\theta - \frac{1}{2}\sin(2\theta)\right) + \left(\phi - \frac{1}{2}\sin(2\phi)\right)\right)\right)^{N_{\rm c}}, & r_{\rm TN} - x_0 < r_0 \le r_{\rm TN} + x_0, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

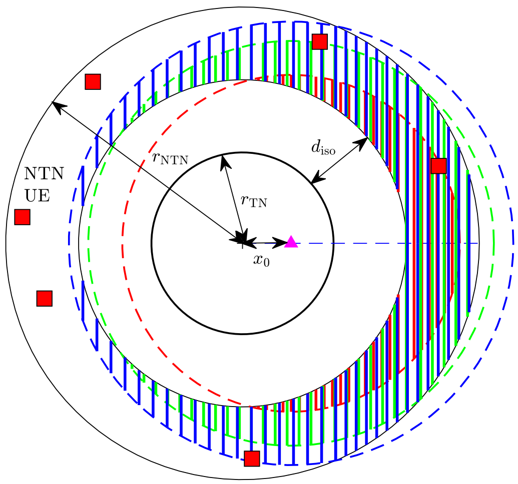


Fig. 2. Geometry of the system model used for derivation of (6) in co-existence Case II. The shaded area is the area obtained from the intersection of the disc centered at TN UE with radius $\mathcal{R}_n$ and the outer annulus defining the outer border for NTN UEs.

$a^2 \tan^{-1}\left(\frac{y-b}{\sqrt{a^2-(b-y)^2}}\right)$. As can be seen from (6), as shown at the bottom of the page, the distribution of interfering distances depends on the location of TN user and the geometry of the network in terms of r_{TN}, r_{iso}, and r_{NTN}. The PDF of r_n can be obtained by taking the derivative w.r.t. r_n.

III. Exact Performance Analysis

In this section, we focus on the performance analysis of the integrated network for the scenarios described in Section II, in terms of coverage probability and data rate of a TN UE which is arbitrarily located on Earth. We utilize stochastic geometry in order to formulate coverage probability and rate as a function of the network's parameters and the propagation characteristics of the channels. One main component of our analytical derivations is the Laplace function of interference which will be presented throughout this section for both cases.

A. Performance Analysis for Co-Existence Case I: NTN DL as the Aggressor and TN DL as the Victim

We define the coverage probability as the probability that the SINR at the TN UE's receiver is greater than a minimum required SINR, denoted by T. Thus, we have

$$P_{\mathrm{c}}(T) \triangleq \mathbb{P}\left(\mathrm{SINR}_{\mathrm{TN}} > T\right). \tag{7}$$

Based on the definition given above, in the following theorem, we derive an analytical expression for the probability of coverage for co-existence case I over a Nakagami-m fading serving channel.

Theorem 1: The coverage probability for a TN user in co-existence Case I is

$$P_{\mathrm{c}}(T) = \int_0^{r_{\mathrm{TN}}+x_0} f_{R_0}(r_0) \Bigg[e^{-s\sigma^2} \sum_{k=0}^{m-1} \frac{\sum_{l=0}^{k} \binom{k}{l} \left(s\sigma^2\right)^l (-s)^{k-l} \frac{\partial^{k-l}}{\partial s^{k-l}} \left(\mathcal{L}_{I_{\mathrm{TN}}}(s) \mathcal{L}_{I_{\mathrm{DL}}}(s)\right)}{k!} \Bigg]_{s=s_{\mathrm{c}}} dr_0, \tag{8}$$

where $s_{\mathrm{c}} = \frac{mTr_0^{\alpha_{\mathrm{TN}}}}{p_{\mathrm{TN}} G_{\mathrm{TN}}}$, $\mathcal{L}_{I_{\mathrm{TN}}}(s)$ and $\mathcal{L}_{I_{\mathrm{DL}}}(s)$ are the Laplace transform of cumulative interference power I_{TN} and I_{DL} which are expressed in Lemma 1 and Lemma 2, respectively.

Proof: See Appendix A. □

Although the final expression involves numerical integration, stochastic geometry formulation eliminates the need to repeatedly sample node locations and channel realizations by replacing a high-dimensional Monte-Carlo simulation with a low-dimensional integral whose complexity does not grow with network size.

Lemma 1: Laplace function of cumulative interference power I_{TN} under general fading channels is

$$\mathcal{L}_{I_{\mathrm{TN}}}(s) \triangleq \mathbb{E}_{I_{\mathrm{TN}}}\left[e^{-sI_{\mathrm{TN}}}\right] = \left(\int_{r_0}^{r_{\mathrm{TN}}+x_0} \mathcal{L}_{H_{\mathrm{TN}}}\left(s p_{\mathrm{TN}} G_{\mathrm{TN}} r_n^{-\alpha_{\mathrm{TN}}}\right) f_{R_n|R_0}(r_n|r_0)\, dr_n\right)^{N_{\mathrm{c}}}. \tag{9}$$

$$F_{r_n}(r_n) = \frac{1}{\pi(r_{\mathrm{NTN}}^2 - (r_{\mathrm{TN}}+r_{\mathrm{iso}})^2)} \times \begin{cases} 0, & 0 \le r_n \le w - x_0, \\ F_{w,0}\left(\frac{w^2 - {r_n}^2 + x_0^2}{2x_0}\right) - F_{r_n,x_0}\left(\frac{w^2 - {r_n}^2 + x_0^2}{2x_0}\right) + F_{r_n,x_0}(x_0 + r_n) - F_{w,0}(w), & w - x_0 < r_n \le r_{\mathrm{NTN}} - x_0, \\ F_{w,0}\left(\frac{w^2 - {r_n}^2 + x_0^2}{2x_0}\right) - F_{r_n,x_0}\left(\frac{w^2 - {r_n}^2 + x_0^2}{2x_0}\right) + F_{r_n,x_0}\left(\frac{r_{\mathrm{NTN}}^2 - {r_n}^2 + x_0^2}{2x_0}\right) + & r_{\mathrm{NTN}} - x_0 < r_n \le w + x_0, \\ F_{r_{\mathrm{NTN}},0}(r_{\mathrm{NTN}}) - F_{w,0}(w) - F_{r_{\mathrm{NTN}},0}\left(\frac{r_{\mathrm{NTN}}^2 - {r_n}^2 + x_0^2}{2x_0}\right), & \\ \pi\left({r_n}^2 - w^2\right) - F_{r_{\mathrm{NTN}},0}(r_{\mathrm{NTN}}) - F_{r_n,x_0}(x_0 + r_n) + & w + x_0 \le r_n \le r_{\mathrm{NTN}} + x_0, \\ F_{r_n,x_0}\left(\frac{r_{\mathrm{NTN}}^2 - {r_n}^2 + x_0^2}{2x_0}\right) + F_{r_{\mathrm{NTN}},0}\left(\frac{r_{\mathrm{NTN}}^2 - {r_n}^2 + x_0^2}{2x_0}\right), & \\ 1, & \text{otherwise}. \end{cases} \tag{6}$$

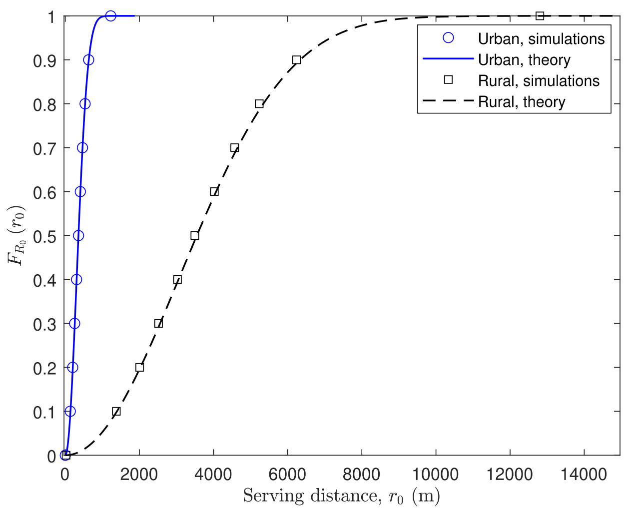


Fig. 3. Verification of serving distance distribution given in (3) for urban (d_{ISD} = 0.75 km) and rural (d_{ISD} = 7.5 km) areas.

Proof: See Appendix B □

Lemma 2: Laplace function of cumulative interference power I_{DL} under general fading channels is

$$\mathcal{L}_{I_{\mathrm{DL}}}(s) \triangleq \mathbb{E}_{\mathscr{R}_n, I_{\mathrm{DL}}}\left[e^{-sI_{\mathrm{DL}}}\right] = \int_a^{r_{\max}} \mathbb{E}_{H_{\mathrm{NTN}}}\left[\exp\left(-sp_{\mathrm{NTN}}G_{\mathrm{NTN}}H_{\mathrm{NTN}}r_n^{-\alpha_{\mathrm{NTN}}}\right)\right] f_{\mathscr{R}_n}(r_n)\, dr_n = \int_a^{r_{\max}} \mathcal{L}_{H_{\mathrm{NTN}}}\left(sp_{\mathrm{NTN}}G_{\mathrm{NTN}}r_n^{-\alpha_{\mathrm{NTN}}}\right) f_{\mathscr{R}_n}(r_n)\, dr_n, \tag{10}$$

where $f_{\mathscr{R}_n}(r_n)$ is the PDF of the distance between the satellite and a TN user which can be obtained from [40, Lemma 1], and $r_{\max} = \sqrt{2r_{\oplus}a + a^2}$ denotes the maximum possible distance between a satellite and a UE that is realized when the satellite is at the user's horizon.

Proof: The Laplace function is obtained by taking expectations over the two random variables, i.e., the distance between the LEO satellite and the TN UE, $\mathscr{R}_n$, and the channel gain H_{NTN}. □

Therefore, the Laplace transform of NTN interference for any fading distribution can be calculated using Lemma 2. Assuming some specific fading distributions, $\mathcal{L}_{H_{\mathrm{NTN}}}(\cdot)$ can be specified at the point $sp_{\mathrm{NTN}}G_{\mathrm{NTN}}r_n^{-\alpha_{\mathrm{NTN}}}$. For instance, when H_{NTN} has a gamma distribution,[1] $\mathcal{L}_{H_{\mathrm{NTN}}}(sp_{\mathrm{NTN}}G_{\mathrm{NTN}}r_n^{-\alpha_{\mathrm{NTN}}}) = \frac{m^m}{(m+sp_{\mathrm{NTN}}G_{\mathrm{NTN}}r_n^{-\alpha_{\mathrm{NTN}}})^m}$, where m is the Nakagami-m fading parameter. Assuming the LEO satellite to be in the zenith of the user, the Laplace function in Lemma 2 will be further simplified to $\mathcal{L}_{I_{\mathrm{DL}}}(s) = \frac{m^m}{(m+sp_{\mathrm{NTN}}G_{\mathrm{NTN}}\,a^{-\alpha_{\mathrm{NTN}}})^m}$.

Corollary 1: When all TN BSs in a cluster transmit over fully orthogonal frequency channels, the interference from TN network will be zero. Thus, we have $I_{\mathrm{TN}} = 0$. Since $a \gg r_{\mathrm{TN}}$, the distance from the interfering satellite to the TN UE can be approximated as $\mathscr{R}_n \approx a$. In this case, the coverage probability for a TN user in co-existence Case I can be obtained in closed-form as

$$P_{\mathrm{c}}(T) = e^{-s\left(\sigma^2 + p_{\mathrm{NTN}}G_{\mathrm{NTN}}a^{-\alpha_{\mathrm{NTN}}}\right)}\left(F_{R_0}(r_{\mathrm{TN}} + x_0) - F_{R_0}(0)\right), \tag{11}$$

where F_{R_0} is given in (3). While it is intuitively clear that increasing the number of BSs enhances coverage for scenario described in Corollary 1, this can also be inferred from (11). As $f_{R_0}(r_0) = \frac{\partial F_{R_0}(r_0)}{\partial r_0}$ becomes larger, the second term in (11) yields a larger value which results in higher coverage probability. The same reasoning applies to d_{ISD}, but in this case the effect is adverse. As can be seen in Fig. 3, the slope of $F_{R_0}(r_0)$ becomes smaller with a larger d_{ISD}, leading to a reduced contribution from the second term in (11) and, hence, a smaller coverage probability.

The average achievable data rate (in bits/s/Hz) is defined as the ergodic capacity for a fading communication link which is derived from Shannon-Hartley theorem and normalized to unit bandwidth. The average achievable rate is defined as

$$\bar{C} \triangleq \mathbb{E}\left[\log_2\left(1 + \mathrm{SINR}_{\mathrm{TN}}\right)\right]. \tag{12}$$

In the following theorem, we derive the expression for the average rate of the victim TN user over Nakagami-m fading serving channel. Both TN and NTN interfering channels may follow any arbitrary distribution.

Theorem 2: The average achievable rate for a TN user in co-existence Case I is

$$\bar{C} = \int_0^{r_{\mathrm{TN}}+x_0}\int_0^{\infty} f_{R_0}(r_0)\left[e^{-s\sigma^2}\sum_{k=0}^{m-1}\frac{\sum_{l=0}^{k}\binom{k}{l}\left(s\sigma^2\right)^l(-s)^{k-l}\frac{\partial^{k-l}}{\partial s^{k-l}}\left(\mathcal{L}_{I_{\mathrm{TN}}}(s)\mathcal{L}_{I_{\mathrm{DL}}}(s)\right)}{k!}\right]_{s=s_{\mathrm{r}}} dt\,dr_0, \tag{13}$$

where $s_{\mathrm{r}} = \frac{m(2^t-1)r_0^{\alpha_{\mathrm{TN}}}}{p_{\mathrm{TN}}G_{\mathrm{TN}}}$, $\mathcal{L}_{I_{\mathrm{TN}}}(s)$ and $\mathcal{L}_{I_{\mathrm{DL}}}$ are obtained from Lemmas 1 and 2, respectively.

Proof: See Appendix C. □

B. Performance Analysis for Co-Existence Case II: NTN UL as the Aggressor and TN DL as the Victim

In the following theorem, we analytically formulate the probability of coverage for co-existence case II, utilizing the definition provided earlier in (7). Similar to Theorem 1, the serving channel follows a Nakagami-m distribution while interfering channels may follow arbitrary distributions.

Theorem 3: The coverage probability for a TN user in co-existence Case II is

$$P_{\mathrm{c}}(T) = \int_0^{r_{\mathrm{TN}}+x_0} f_{R_0}(r_0)\left[e^{-s\sigma^2}\sum_{k=0}^{m-1}\frac{\sum_{l=0}^{k}\binom{k}{l}\left(s\sigma^2\right)^l(-s)^{k-l}\frac{\partial^{k-l}}{\partial s^{k-l}}\left(\mathcal{L}_{I_{\mathrm{TN}}}(s)\mathcal{L}_{I_{\mathrm{UL}}}(s)\right)}{k!}\right]_{s=s_{\mathrm{c}}} dr_0, \tag{14}$$

[1] The gain of the channel, which is the square of Nakagami random variable, follows a gamma distribution.

where $s_{\mathrm{c}} = \frac{mTr_0^{\alpha_{\mathrm{TN}}}}{p_{\mathrm{TN}}G_{\mathrm{TN}}}$, and $\mathcal{L}_{I_{\mathrm{UL}}}(s)$ is the Laplace transform of cumulative interference power I_{UL} from all NTN UEs located on the outer annulus shown in Fig. 2 that is expressed in Lemma 3. Obviously, $\mathcal{L}_{I_{\mathrm{TN}}}(s)$ remains the same as for Case I and is obtained from Lemma 1.

Proof: The expression can be obtained using similar steps as shown in the proof of Theorem 1 given in Appendix A by only substituting $I_{\mathrm{UL}} \rightarrow I_{\mathrm{DL}}$. □

Lemma 3: Laplace function of cumulative interference power I_{UL} under general fading channels is

$$\begin{aligned}\mathcal{L}_{I_{\mathrm{UL}}}(s) &\triangleq \mathbb{E}_{I_{\mathrm{UL}}}\left[e^{-sI_{\mathrm{UL}}}\right]\\ &= \bigg(\int_{w-x_0}^{r_{\mathrm{NTN}}+x_0} \mathcal{L}_{H_{\mathrm{NTN}}}\left(sp_{\mathrm{NTN}}G_{\mathrm{NTN}}r_n^{-\alpha_{\mathrm{TN}}}\right)\\ &\qquad f_{r_n}(r_n)\,dr_n\bigg)^{N_{\mathrm{u}}}. \end{aligned} \tag{15}$$

Proof: See Appendix D □

Corollary 2: When all TN BSs in a cluster transmit over fully orthogonal frequency channels, and also assuming that $r_{\mathrm{iso}} \gg 0$, the system becomes noise-limited and the upper bound for coverage probability of a TN user can be obtained in closed-form as

$$P_{\mathrm{c}}(T) = e^{-s\sigma^2}\left(F_{R_0}\left(r_{\mathrm{TN}} + x0\right) - F_{R_0}(0)\right), \tag{16}$$

where F_{R_0} is given in (3). Similar to Corollary 1, insights can be obtained directly from Corollary 2 that aligns with intuition. The upper bound improves by increasing the number of TN BSs due to the steepening the slope of $F_{R_0}(r_0)$ and enlarging the second term in (16), whereas parameter d_{ISD} has the opposite effect as it flattens the slope and reduces its contribution to (16).

In the following theorem, we formulate the expression for the average rate of the victim TN user over Nakagami-m fading serving channel for co-existence Case II.

Theorem 4: The average achievable rate for a TN user in co-existence Case II is

$$\begin{aligned}\bar{C} = \int_0^{r_{\mathrm{TN}}+x_0}\int_0^{\infty} f_{R_0}(r_0)\Bigg[e^{-s\sigma^2}\\ \sum_{k=0}^{m-1}\frac{\sum_{l=0}^{k}\binom{k}{l}\left(s\sigma^2\right)^l(-s)^{k-l}\frac{\partial^{k-l}}{\partial s^{k-l}}\left(\mathcal{L}_{I_{\mathrm{TN}}}(s)\mathcal{L}_{I_{\mathrm{UL}}}(s)\right)}{k!}\Bigg]_{s=s_{\mathrm{r}}} dt\,dr_0,\end{aligned} \tag{17}$$

where $s_{\mathrm{r}} = \frac{m\left(2^t-1\right)r_0^{\alpha_{\mathrm{TN}}}}{p_{\mathrm{TN}}G_{\mathrm{TN}}}$. $\mathcal{L}_{I_{\mathrm{TN}}}(s)$ and $\mathcal{L}_{I_{\mathrm{UL}}}(s)$ are obtained from Lemmas 1 and 3, respectively.

Proof: The proof follows similar procedure as shown in the proof of Theorem 2 presented in Appendix C. The sole difference is the origin of NTN cumulative interference power, which, in this case, arises from NTN UL. □

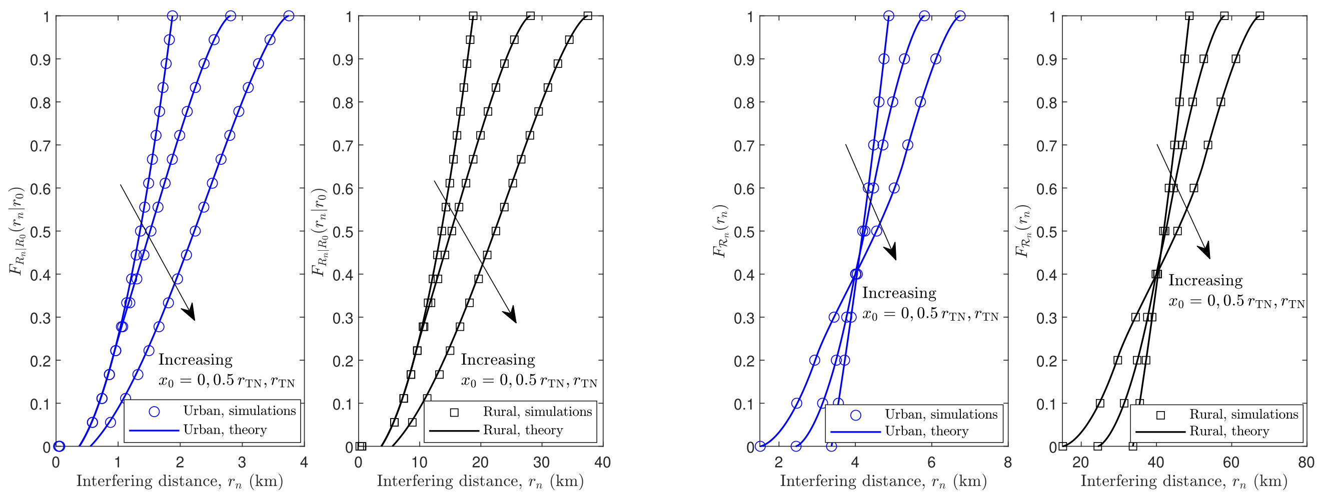
IV. Numerical Results

This section focuses on the numerical evaluation of the analytical expressions, from which we obtain insights on the performance and parameter sensitivity of the integrated network. Monte-Carlo simulations are provided solely to confirm the accuracy of the analytical derivations, rather than to generate the insights themselves. In this section, analytical derivations on the performance of the integrated network for each case are verified using Monte Carlo simulations. The Monte Carlo results are based on 250000 independent realizations with distinct random seeds. Confidence intervals of 95% were calculated but omitted from the plots for clarity. In addition, several insights on the performance of the integrated network can be inferred from the numerical results. The TN BSs and satellites antenna models are chosen according to the sectorized antenna patterns given in [4] for simulations. However, in order to maintain tractability for the theoretical results, we approximate such model by assuming fixed antenna gains for the main and side lobes of the antennas, similar to [52], [53], and [54]. This simplified approach is sufficient to capture the generic trends for system-level interference analysis and feasibility studies without introducing unnecessary complexity from angular dependencies. In fact, the main beam transmits with the maximum antenna gain given in [4], while side lobes are assumed to have 13 dB lower gain, similar to [51] and [52]. The 3 dB beamwidth and maximum gain of satellites' antennas are 4.4127° and 30 dBi, respectively. For TN BSs, the maximum antenna gain is 17 dBi and mechanical down-tilt angles are 3° and 10°, respectively, for rural and urban areas [4]. Horizontal 3 dB beamwidth of antenna element are 90°. Vertical 3 dB beamwidths are 54° and 65° for urban and rural areas, respectively. Both TN and NTN UEs are equipped with omni-directional antennas with gain of 0 dBi as in [4].

We consider the path-loss exponents to be $\alpha_{\mathrm{TN}} = 3$ and $\alpha_{\mathrm{NTN}} = 2$ for TN and NTN propagation channels, respectively, according to [55, Table 3.2]. We adopt a practical fading model, i.e., Rician, for all simulation results with parameter K, where $K = 0$ and 200 for TN and NTN channels, respectively, to account for higher LoS in NTN. Since the theoretical derivations are developed assuming a Nakagami-m fading for the serving link to preserve the analytical tractability, a Nakagami-m distribution is used for their numerical calculations by setting its parameter to $m = \frac{k^2+2K+1}{2K+1}$, by matching the first and second moments of Rician and Nakagami PDFs. The fair match between theoretical results and simulations confirms the high accuracy of such approximation.

Satellite transmit power, cell radius, maximum antenna gain are $p_{\mathrm{t}} = 46$ dBm, 25 km, and 30 dBi, respectively. The main lobe gain of the BSs antennas is assumed to be 17 dBi. The BS transmit power is assumed to be $p_{\mathrm{t}} = 46$ dBm. The altitude is set to $\{200, 600, 1200\}$ km, unless stated otherwise. TN inter-site distances are assumed to be 0.75 km and 7.5 km for urban and rural areas, respectively. The number of BSs on each TN cluster is assumed to be 19. The operating frequency and bandwidth are set to be 2 GHz and 20 MHz, respectively. It should be noted that, in this study, we kept the per beam effective isotropic radiated power (EIRP) constant to isolate geometric and altitude-dependent effects, establishing a conservative upper bound on altitude sensitivity. Real-world systems typically adjust the transmit power based on altitude, beam pointing, latitude, and service usage, as reflected in the ITU power flux density and EIRP requirements in Article 22 of the RR.

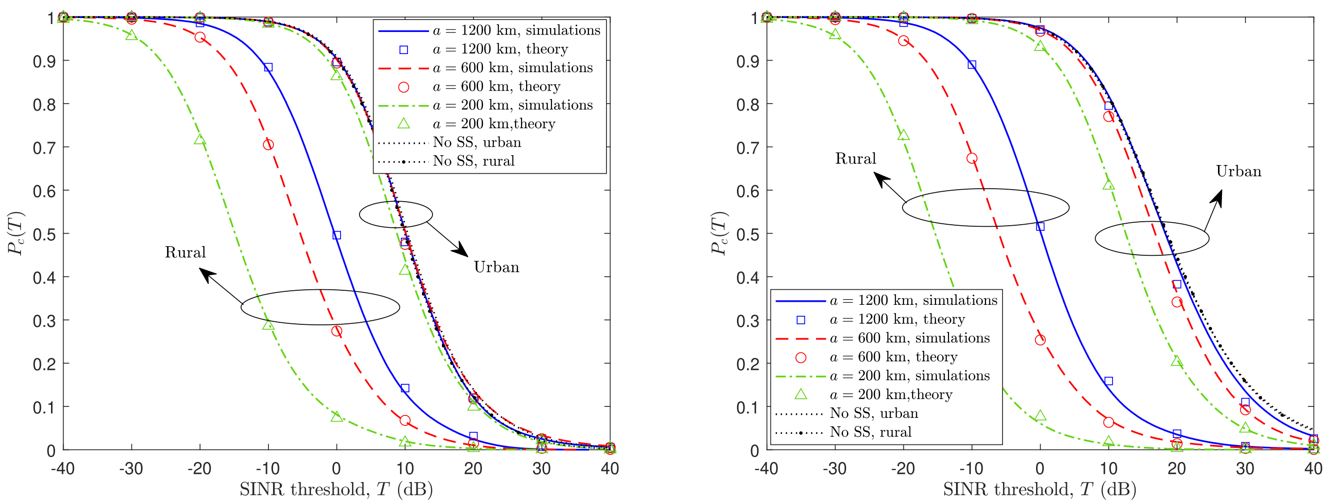
Figure 3 verifies the derivation of serving distance distribution given in (3) for urban and rural areas. It is worth noting



(a) Case I.

(b) Case II.

Fig. 4. Verification of the interfering distance distribution, for interference caused by TN network (a), given as in (4), and interference caused by NTN UEs (b) given as in (6).



(a) 100% load.

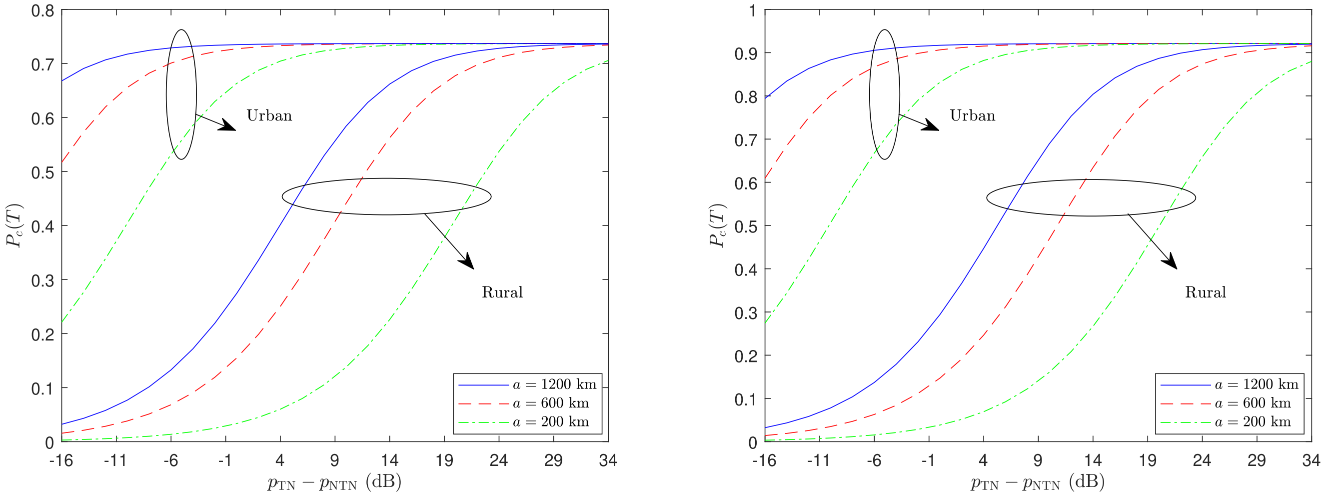
(b) 25% load.

Fig. 5. Coverage probability of Case I for 100% load (a), and 25% load (b), for different LEO satellite altitudes. The simulations, shown by lines, verify the theoretical results, depicted by markers, given in Theorem 1.

that the serving distance is the same for both Cases I and II. Figures 4(a) and 4(b) verify the CDF of interference from TN BSs and NTN UEs derived in (4) and (6), respectively, in rural and urban areas. The results are presented for a TN user located in the center, middle, and the edge of the TN cluster, i.e., $x_0 = \{0, 0.5\,r_{\mathrm{TN}}, r_{\mathrm{TN}}\}$. As can be seen in the figures 3 and 4, there is a perfect match between the analytical results, shown by lines, and the simulations, shown by markers, which verifies the exactness of the derivations.

Figures 5(a) and 5(b) verify the derivations in Theorem 1 for 100% and 25% load from the TN network. The load level represents what percentage of BSs are active within the TN cluster. The black dotted lines represent the case of having no spectrum sharing between TN and NTN, i.e., the whole band is allocated only to TN. We can observe that the performance degradation due to the sharing of the spectrum is less significant in urban areas. The reason is that BSs are located in relatively closer proximity, i.e., smaller d_{ISD}, in urban areas, so that the interference received from TN BSs is much larger than the interference received from the LEO satellite, especially for 100% load. On the other hand, for rural network, as the BSs are much farther away from each other, NTN interference becomes more comparable to TN interference. Thus, there is more performance degradation for rural areas when sharing the spectrum between TN and NTN. Obviously, the impact of the LEO satellite altitude is also more significant in rural regions.

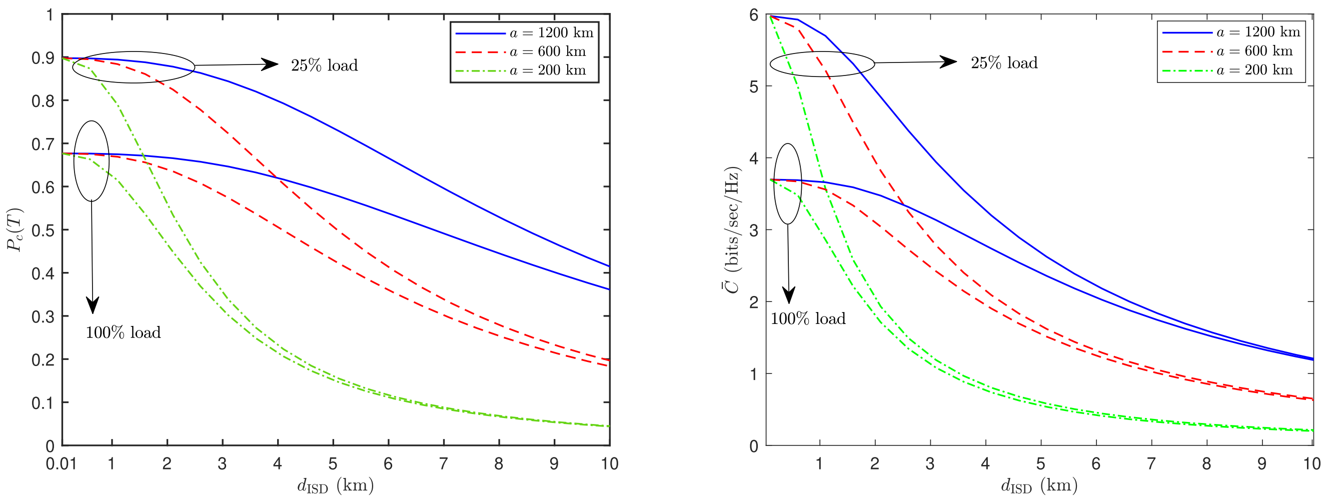
Figures 6(a) and 6(b) show the effect of the power allocation between TN and NTN on the coverage probability. The transmit power of the LEO satellite is assumed to be fixed and equal to $p_{\mathrm{NTN}} = 46$ dBm. By increasing the TN power, the coverage probability saturates to a fixed value for which the NTN has no effect on its performance. The saturation value depends on the number of interfering TN BSs. As can be seen



(a) 100% load.

(b) 25% load.

Fig. 6. Effect of transmit power on coverage probability of Case I for 100% load (a), and 25% load (b).



(a) Coverage probability

(b) Data rate

Fig. 7. Effect of d_{ISD} on Coverage probability (a) and data rate (b) of Case I for different LEO satellite altitudes and TN load levels.

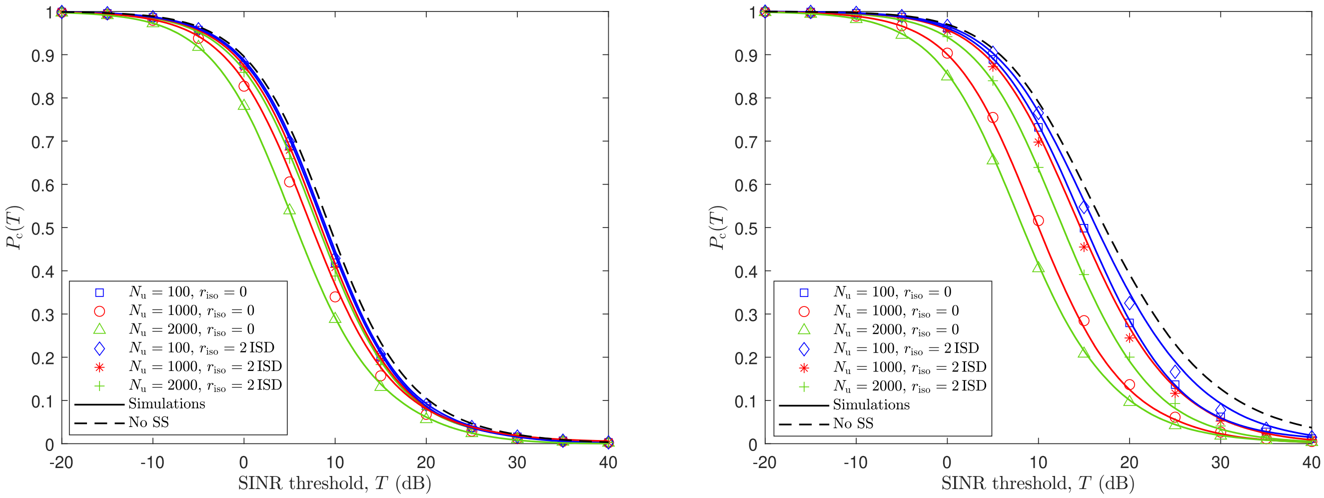
in the figure, a higher TN power is required for larger d_{ISD}, i.e., rural regions, as well as for lower satellite altitudes to maximize the coverage probability.

The effect of d_{ISD} for different satellite altitudes and TN load levels on the coverage probability and the data rate are illustrated in Figs. 7(a) and 7(b), respectively. Lower d_{ISD} provides better coverage and rate due to more strength of the serving signal. The performance drops with increasing the d_{ISD}. The decrease is more considerable for lower LEO altitudes due to higher interference power.

For Case II, the transmit power of NTN UEs is assumed to be 200 mW. Figures 8(a) and 8(b) verify the derivations in Theorem 3 for 100% and 25% load from the TN cell and varying the number of NTN UEs and isolation distances. The results show that the interference from TN DL is more significant compared to NTN UL, especially for higher loads. The effect of spectrum sharing is rather minimal for 100% load from TN network, especially when isolation distance is greater than $2d_{\mathrm{ISD}}$.

The effect of the isolation distance between the TN and NTN networks is shown in Fig. 9, for $a = 200$ km. As can be seen in the figure, increasing the isolation distance has no significant impact on the performance of users located at the center or middle of the TN cell. For edge users, a larger isolation distance is needed to achieve the highest possible performance. The minimum required isolation distance also decreases with increasing TN load level since NTN interference becomes less comparable to TN interference.

Figure 10 illustrates the coverage probability for both coexistence scenarios as a function of the total number of TN BSs in a cluster. According to the figure, for most cases, increasing the number of BSs initially improves the coverage probability, but it eventually saturates at an upper bound determined by system parameters. For edge UE in Case II with $r_{\mathrm{iso}} = 2d_{\mathrm{ISD}}$, the coverage is significantly higher than in Case I for a smaller number of BSs. However, as the number of BSs becomes large, the coverage saturates to nearly



(a) 100% load.

(b) 25% load.

Fig. 8. Coverage probability of Case II for 100% load (a), and 25% load (b) for different number of NTN UEs and isolation distances. The simulations, shown by lines, verify the theoretical results, depicted by markers, given in Theorem 3.

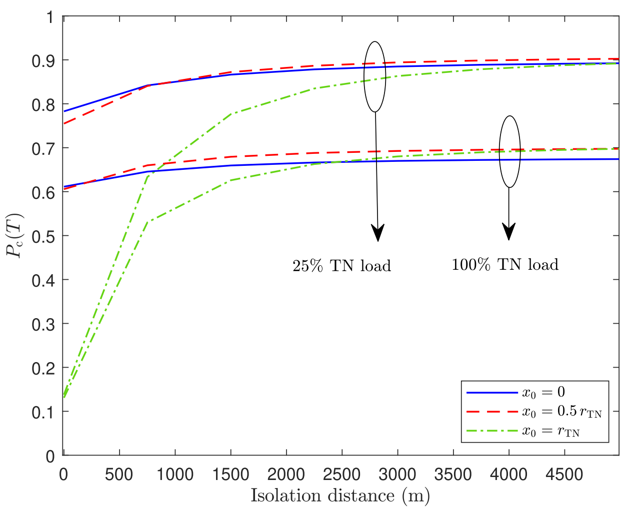


Fig. 9. Effect of isolation distance, r_{iso}, on coverage probability of Case II for different TN UE's locations.

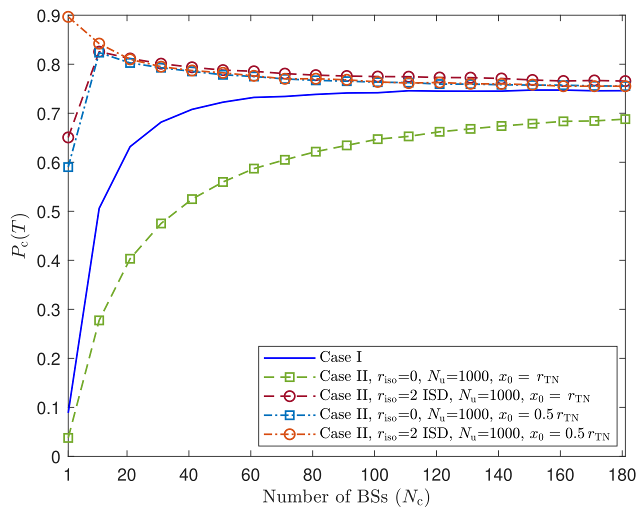


Fig. 10. Effect of TN BS density on coverage probability of Case I and II.

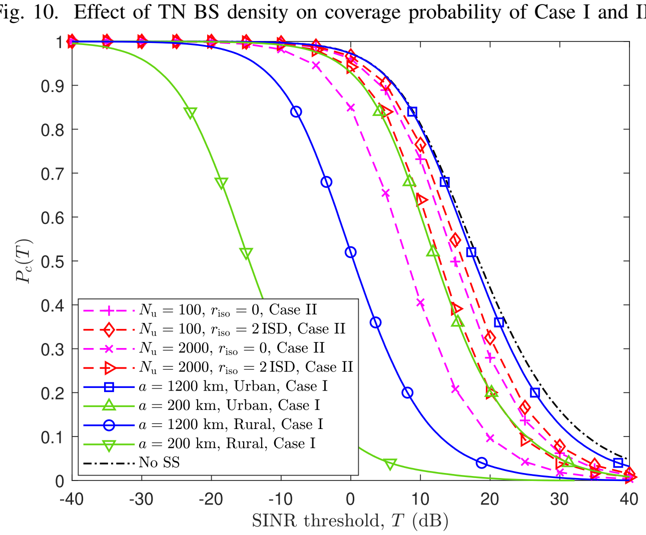


Fig. 11. Comparison of the two co-existence cases studied in this paper.

the same upper limit as in Case I. Conversely, the edge UE in Case II with zero isolation distance demonstrates the poorest performance among the scenarios and reaches its upper limit more slowly. This is because the UE is located too close to interfering NTN UEs transmitting in the uplink. The middle UE in Case II with $r_{\mathrm{iso}} = 2d_{\mathrm{ISD}}$ achieves the best performance, with faster saturation as the BS density increases. The middle UE benefits from being farther away from NTN interfering UEs, which adds to the isolation distance, reducing interference from NTN UEs and resulting in a higher coverage probability.

In Fig. 11, we directly compare the two co-existence cases. As can be seen in this figure, one co-existence case may outperform the other depending on several system parameters, e.g., the number of interfering NTN UEs, isolation distance, altitude of the constellation, density of TN BSs (urban or rural). It is worth noting that the d_{ISD} has no considerable effect on the performance in Case II. Thus, the results shown for Case II hold for both rural and urban areas. Based on these results, different guidelines can be provided for the network to switch between the two cases. For rural area, co-existence Case II always provides superior performance regardless of the altitude of the satellite. In urban regions, if the altitude of the satellite is large (1200 km as an example in Fig. 11), Case I outperforms Case II regardless of number of interfering NTN UEs or isolation distance. For urban regions, if r_{iso} is small and N_{u} is large (2000 as an example in Fig. 11), Case I will outperform Case II.

V. Conclusion

In this paper, we studied the performance of the integrated TN-NTN network when TN and NTN operate on the same frequency channel in the S-band. Two co-existence cases were considered, namely Case I and II, where TN users receive interference from NTN downlink and uplink, respectively. Using tools from stochastic geometry, we obtained exact tractable expressions on the coverage probability and average data rate for each case in terms of the TN and NTN networks' parameters. The analysis and results presented herein offer key insights into optimal dynamic spectrum sharing between TN and NTN by tuning different system parameters. In particular, for Case I, the findings reveal that for larger TN inter-site-distance and/or lower satellites' altitude, a higher TN power is required to improve the performance. For Case II, the effect of spectrum sharing is rather minimal when there is a sufficient isolation distance, especially when TN load level is high. Moreover, the minimum required isolation distance is determined based on the required performance for edge users. Case II with a sufficient isolation distance provides better performance for edge users compared to Case I when the BS density is low. For higher BS density, both cases saturate to nearly the same upper limit.

The study showed that, depending on the network parameters, one co-existence case may outperform the other. Thus, the network may dynamically switch between the two cases to optimize the performance. In particular, For rural area, co-existence Case II always provides superior performance regardless of the altitude of the satellite. In urban regions, if altitude of satellite is large, Case I outperforms Case II regardless of number of interfering NTN UEs or isolation distance. For urban regions, if isolation distance is small and the number of NTN UEs is large, Case I will outperform Case II.

Appendix

A. Proof of Theorem 1

Let us start derivation of Theorem 1 using the definition of coverage probability given in (7):

$$
\begin{aligned}
&P_{\mathrm{c}}(T) = \mathbb{P}(\mathrm{SINR}_{\mathrm{TN}} > T) = \mathbb{E}_{R_0}\left[\mathbb{P}\left(\mathrm{SINR}_{\mathrm{TN}} > T | R_0 = r_0\right)\right] \\
&= \int_0^u \mathbb{P}\left(\mathrm{SINR}_{\mathrm{TN}} > T | R_0 = r_0\right) f_{R_0}(r_0)\, dr_0 \\
&= \int_0^u \mathbb{P}\left(\frac{p_{\mathrm{TN}} G_{\mathrm{TN}} H_{\mathrm{TN}} r_0^{-\alpha_{\mathrm{TN}}}}{I_{\mathrm{TN}} + I_{\mathrm{DL}} + \sigma^2} > T\right) f_{R_0}(r_0)\, dr_0 \\
&= \int_0^u \mathbb{E}_{I_{\mathrm{TN}}, I_{\mathrm{DL}}}\left[\mathbb{P}\left(H_{\mathrm{TN}} > \frac{T\left(I_{\mathrm{TN}} + I_{\mathrm{DL}} + \sigma^2\right) r_0^{\alpha_{\mathrm{TN}}}}{p_{\mathrm{TN}} G_{\mathrm{TN}}}\right) \middle| I_{\{\mathrm{TN,DL}\}} > 0\right] \\
&\times f_{R_0}(r_0)\, dr_0 \\
&= \int_0^u \mathbb{E}_{I_{\mathrm{TN}}, I_{\mathrm{DL}}}\left[1 - F_{H_{\mathrm{TN}}}\left(\frac{T\left(I_{\mathrm{TN}} + I_{\mathrm{DL}} + \sigma^2\right) r_0^{\alpha_{\mathrm{TN}}}}{p_{\mathrm{TN}} G_{\mathrm{TN}}}\right)\right] \\
&\times f_{R_0}(r_0)\, dr_0 \\
&\overset{(a)}{=} \int_0^u \mathbb{E}_{I_{\mathrm{TN}}, I_{\mathrm{DL}}}\left[\frac{\Gamma\left(m, m\frac{T\left(I_{\mathrm{TN}} + I_{\mathrm{UL}} + \sigma^2\right) r_0^{\alpha_{\mathrm{TN}}}}{p_{\mathrm{TN}} G_{\mathrm{TN}}}\right)}{\Gamma(m)}\right] f_{R_0}(r_0)\, dr_0 \\
&\overset{(b)}{=} \int_0^u f_{R_0}(r_0)\, e^{-\frac{m T \sigma^2 r_0^{\alpha_{\mathrm{TN}}}}{p_{\mathrm{TN}} G_{\mathrm{TN}}}} \mathbb{E}_{I_{\mathrm{TN}}, I_{\mathrm{DL}}}\Bigg[e^{-m\frac{T\left(I_{\mathrm{TN}} + I_{\mathrm{DL}}\right) r_0^{\alpha_{\mathrm{TN}}}}{p_{\mathrm{TN}} G_{\mathrm{TN}}}} \\
&\sum_{k=0}^{m-1} \frac{\sum_{l=0}^{k}\binom{k}{l}\left(m\frac{T\sigma^2 r_0^{\alpha_{\mathrm{TN}}}}{p_{\mathrm{TN}} G_{\mathrm{TN}}}\right)^l\left(m\frac{T\left(I_{\mathrm{TN}} + I_{\mathrm{DL}}\right) r_0^{\alpha_{\mathrm{TN}}}}{p_{\mathrm{TN}} G_{\mathrm{TN}}}\right)^{k-l}}{k!}\Bigg] dr_0 \\
&\overset{(c)}{=} \int_0^u f_{R_0}(r_0)\Bigg[e^{-s\sigma^2} \\
&\sum_{k=0}^{m-1} \frac{\sum_{l=0}^{k}\binom{k}{l}\left(s\sigma^2\right)^l(-s)^{k-l}\frac{\partial^{k-l}}{\partial s^{k-l}}\mathcal{L}_{I_{\mathrm{TN}}}(s)\mathcal{L}_{I_{\mathrm{DL}}}(s)}{k!}\Bigg]_{s=s_0} dr_0,
\end{aligned}
\tag{18}
$$

where $u = r_{\mathrm{TN}} + x_0$, $F_{H_{\mathrm{TN}}}(\cdot)$ is the CDF of H_{TN}, (a) follows from the distribution of gamma random variable H_{TN} (being the square of the Nakagami random variable), (b) is calculated by applying the definition of incomplete gamma function for integer values of m to (a), and (c) follows from independence of I_{TN} and I_{NTN}, i.e., $\mathbb{E}_{I_{\mathrm{NTN}}, I_{\mathrm{TN}}}\left[e^{-s(I_{\mathrm{NTN}} + I_{\mathrm{TN}})}\right] = \mathbb{E}_{I_{\mathrm{NTN}}}\left[e^{-sI_{\mathrm{NTN}}}\right]\mathbb{E}_{I_{\mathrm{TN}}}\left[e^{-sI_{\mathrm{TN}}}\right]$.

B. Proof of Lemma 1

Using the definition of the Laplace transform yields

$$
\begin{aligned}
&\mathcal{L}_{I_{\mathrm{TN}}}(s) \triangleq \mathbb{E}_{I_{\mathrm{TN}}}\left[e^{-sI_{\mathrm{TN}}}\right] \\
&= \mathbb{E}_{R_n, H_{\mathrm{TN}}}\left[\exp\left(-s\sum_{n=1}^{N_{\mathrm{c}}} p_{\mathrm{TN}} G_{\mathrm{TN}} H_{\mathrm{TN}} R_n^{-\alpha_{\mathrm{TN}}}\right)\right] \\
&= \mathbb{E}_{R_n, H_{\mathrm{TN}}}\left[\prod_{n=1}^{N_{\mathrm{c}}} \exp\left(-s p_{\mathrm{TN}} G_{\mathrm{TN}} H_{\mathrm{TN}} R_n^{-\alpha_{\mathrm{TN}}}\right)\right] \\
&\overset{(a)}{=} \mathbb{E}_{R_n}\left[\prod_{n=1}^{N_{\mathrm{c}}} \mathbb{E}_{H_{\mathrm{TN}}}\left[\exp\left(-s p_{\mathrm{TN}} G_{\mathrm{TN}} H_{\mathrm{TN}} R_n^{-\alpha_{\mathrm{TN}}}\right)\right]\right] \\
&\overset{(b)}{=} \prod_{n=1}^{N_{\mathrm{c}}} \int_{r_0}^{u} \mathbb{E}_{H_{\mathrm{TN}}}\left[\exp\left(-s p_{\mathrm{TN}} G_{\mathrm{TN}} H_{\mathrm{TN}} r_n^{-\alpha_{\mathrm{TN}}}\right)\right] \\
&\times f_{R_n|R_0}(r_n|r_0)\, dr_n \\
&\overset{(c)}{=} \left(\int_{r_0}^{u} \mathcal{L}_{H_{\mathrm{TN}}}\left(s p_{\mathrm{TN}} G_{\mathrm{TN}} r_n^{-\alpha_{\mathrm{TN}}}\right) f_{R_n|R_0}(r_n|r_0)\, dr_n\right)^{N_{\mathrm{c}}},
\end{aligned}
\tag{19}
$$

where $u = r_{\mathrm{TN}} + x_0$, (a) follows from the i.i.d. distribution of H_{TN} and its further independence from R_n, (b) is obtained using the interfering distance distribution from (5), and (c) is derived thanks to the i.i.d. distribution of conditional PDF of $f_{R_n|R_0}(r_n|r_0)$ and the substitution from the definition of Laplace function.

C. Proof of Theorem 2

Based on the definition of the average data rate given in (12), we have

$$\begin{aligned}\bar{C} &= \mathbb{E}_{I_{\rm TN},I_{\rm DL},H_{\rm TN},R_0}[\log_2(1+\mathrm{SINR}_{\rm TN})]\\ &= \int_0^u \mathbb{E}_{I_{\rm TN},I_{\rm DL},H_{\rm TN}}\left[\log_2(1+\mathrm{SINR}_{\rm TN})\right] f_{R_0}(r_0)\,dr_0\\ &\overset{(a)}{=} \int_0^u\int_0^\infty \mathbb{E}_{I_{\rm TN},I_{\rm DL},H_{\rm TN}}\left[\mathbb{P}\left(\log_2(1+\mathrm{SINR}_{\rm TN})>t\right)\right]\\ &\quad f_{R_0}(r_0)\,dtdr_0,\end{aligned} \tag{20}$$

where $u=r_{\rm TN}+x_0$, $\mathrm{SINR}_{\rm TN}=\frac{p_{\rm TN}G_{\rm TN}H_{\rm TN}r_0^{-\alpha_{\rm TN}}}{I_{\rm TN}+I_{\rm DL}+\sigma^2}$, and (a) is obtained from the fact that for a positive random variable X, $\mathbb{E}[X]=\int_{t>0}\mathbb{P}(X>t)\,dt$. Thus, we have

$$\begin{aligned}\bar{C} &\overset{(a)}{=}\\ &\int_0^u\int_0^\infty \mathbb{E}_{I_{\rm TN},I_{\rm DL}}\left[1-F_{H_{\rm TN}}\left(\frac{r_0^{\alpha_{\rm TN}}\left(I_{\rm TN}+I_{\rm DL}+\sigma^2\right)(2^t-1)}{p_{\rm TN}G_{\rm TN}}\right)\right]\\ &\times f_{R_0}(r_0)\,dt\;dr_0\\ &\overset{(b)}{=}\int_0^u\int_0^\infty \mathbb{E}_{I_{\rm TN},I_{\rm DL}}\left[\frac{\Gamma\left(m, m\frac{r_0^{\alpha_{\rm TN}}\left(I_{\rm TN}+I_{\rm DL}+\sigma^2\right)(2^t-1)}{p_{\rm TN}G_{\rm TN}}\right)}{\Gamma(m)}\right]\\ &\times f_{R_0}(r_0)\,dt\;dr_0\\ &\overset{(c)}{=}\int_0^u\int_0^\infty f_{R_0}(r_0)e^{-\frac{m\,\sigma^2(2^t-1)r_0^{\alpha_{\rm TN}}}{p_{\rm TN}G_{\rm TN}}}\mathbb{E}_{I_{\rm TN},I_{\rm DL}}\Bigg[e^{-m\frac{(I_{\rm DL}+I_{\rm TN})r_0^{\alpha_{\rm TN}}}{p_{\rm TN}G_{\rm TN}}}\\ &\sum_{k=0}^{m-1}\frac{\sum_{l=0}^{k}\binom{k}{l}\left(m\frac{\sigma^2(2^t-1)r_0^{\alpha_{\rm TN}}}{p_{\rm TN}G_{\rm TN}}\right)^l\left(m\frac{(I_{\rm DL}+I_{\rm TN})r_0^{\alpha_{\rm TN}}}{p_{\rm TN}G_{\rm TN}}\right)^{k-l}}{k!}\Bigg]dtdr_0,\end{aligned} \tag{21}$$

where (a) follows from the product distribution of two independent random variables, (b) follows from the gamma distribution of serving channel gain $H_{\rm TN}$, and (c) is calculated by applying the incomplete gamma function for integer values of m to (b).

D. Proof of Lemma 3

Using the definition of the Laplace transform yields

$$\begin{aligned}\mathcal{L}_{I_{\rm UL}}(s) &\triangleq \mathbb{E}_{I_{\rm UL}}\left[e^{-sI_{\rm UL}}\right]\\ &= \mathbb{E}_{\mathcal{R}_n,H_{\rm NTN}}\left[\exp\left(-s\sum_{n=1}^{N_{\rm u}}p_{\rm NTN}G_{\rm NTN}H_{\rm NTN}\mathcal{R}_n^{-\alpha_{\rm TN}}\right)\right]\\ &= \mathbb{E}_{\mathcal{R}_n,H_{\rm NTN}}\left[\prod_{n=1}^{N_{\rm u}}\exp\left(-sp_{\rm NTN}G_{\rm NTN}H_{\rm NTN}\mathcal{R}_n^{-\alpha_{\rm TN}}\right)\right]\\ &\overset{(a)}{=}\mathbb{E}_{\mathcal{R}_n}\left[\prod_{n=1}^{N_{\rm u}}\mathbb{E}_{H_{\rm NTN}}\left[\exp\left(-sp_{\rm NTN}G_{\rm NTN}H_{\rm NTN}\mathcal{R}_n^{-\alpha_{\rm TN}}\right)\right]\right]\\ &\overset{(b)}{=}\prod_{n=1}^{N_{\rm u}}\int_{w-x_0}^{r_{\rm NTN}+x_0}\mathbb{E}_{H_{\rm NTN}}\left[\exp\left(-sp_{\rm NTN}G_{\rm NTN}H_{\rm NTN}r_n{}^{-\alpha_{\rm TN}}\right)\right]\\ &\qquad\times f_{\mathcal{R}_n}(r_n)\,dr_n\\ &\overset{(c)}{=}\left(\int_{w-x_0}^{r_{\rm NTN}+x_0}\mathcal{L}_{H_{\rm NTN}}\left(sp_{\rm NTN}G_{\rm NTN}r_n^{-\alpha_{\rm TN}}\right)f_{\mathcal{R}_n}(r_n)\,dr_n\right)^{N_{\rm u}},\end{aligned} \tag{22}$$

where $w=r_{\rm TN}+r_{\rm iso}$, (a) follows from the i.i.d. distribution of $H_{\rm NTN}$ and its further independence from $\mathcal{R}_n$, (b) is obtained using the derivative of interfering distance CDF from (6), (c) is derived thanks to the i.i.d. distribution of PDF of $f_{\mathcal{R}_n}(r_n)$ and the substitution from the definition of Laplace function.

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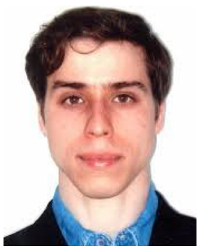

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